

**Protocol: A randomised trial of krill oil for osteoarthritis of the knee
(Protocol number: KAROAKE trial, version 02, February 2016)**

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A randomised trial of krill oil for osteoarthritis of the knee

Background: Knee osteoarthritis (OA) is a significant and increasing cause of pain and disability in the Australian population. So far, interventions aimed at structural modification have been disappointing. OA affects the whole joint, and inflammation of the joint lining (synovitis and joint fluid) is now recognised as a key part of osteoarthritis.

Animal studies suggest that krill oil is anti-inflammatory. Pilot data in humans shows that krill oil reduces knee pain and systemic inflammation after 30 days, suggesting that an approach targeted at specific mechanisms of disease may be effective for OA. However, this study was of short duration, did not collect any data on structural factors, and included patients with a variety of disorders. Longer, larger follow up studies are required.

Aim: To compare, using a randomised, placebo-controlled double-blind design over six months, the efficacy of krill oil vs identical placebo to treat knee osteoarthritis (OA) (both pain and structure) in 260 patients with clinical knee OA, significant knee pain and effusion on imaging.

Primary hypothesis:

Krill oil decreases pain (assessed by 100mm VAS) by 10mm more than identical placebo over 24 weeks in patients with clinical knee OA, significant knee pain and effusions.

Secondary hypotheses:

Krill oil decreases effusion size by 20% over 24 weeks in patients with symptomatic osteoarthritis of the knee and knee effusion.

Krill oil improves knee pain over 4, 8, 12, 16, and 20 weeks in patients with clinical knee OA and effusion.

Significance and Innovation: If krill oil can both improve symptoms and decrease effusion in OA it has the potential to slow progression to joint replacement. The proposed study represents an innovative approach to this and lends itself to easy implementation as krill oil is already available over-the-counter in Australian pharmacies.

Background

Osteoarthritis (OA) is one of the leading causes of disability globally. In Australia, OA affects over 1.9 million people (8.6% of the population – 2012 estimates), with health expenditure of \$3.75 billion in 2012¹. The majority of the cost is hip and knee replacements^{1,2}, with over 40,000 of these procedures performed in 2006–07, each costing an average of \$15,000–\$30,000².

Despite its large disease burden, there are currently no approved disease-modifying drugs available which modify structural progression of OA. Current treatments focus on alleviating pain, which is important because patients rank pain control as their highest treatment priority³. Despite this, pain control remains poor in 50% of patients⁴. Therefore, *there is a significant gap between the level of pain relief patients want, and what they actually get*. Lack of treatment efficacy may be partly due to treating everyone with OA as if they have the same pathological process.

Modern approaches to OA involve targeting specific disease pathways

OA is now considered to be a collection of different disease pathways that result in a common outcome^{5,6}, rather than one disease with a common pathway. There are many sources of pain, and many factors associated with loss of cartilage (Table 1)⁷, and thereby progression of disease. Our approach is that *targeting specific sources of pain and structural damage is likely to improve outcomes in individual patients*. We have successfully used this approach, targeting bone marrow lesions, where we produced the first evidence of a treatment that had a larger effect on pain *and*

modified structure⁸. This Phase II trial has been followed by a multicentre trial testing this approach for cartilage loss and joint replacement (NHMRC Grant 1045415 CIA Jones). As well as bone, inflammation is a strong target in OA.

Table 1: Associations between knee structural factors, pain and cartilage loss⁷

Factor	Independently associated with pain	Associated with cartilage loss
Obesity, muscle strength, bone marrow lesions (BMLs), cartilage defects, meniscal tears, meniscal extrusions, effusion, synovitis	Yes	Yes
Tibial bone size, cartilage volume	No	Yes
Subchondral bone mass	No	Yes ^a
Anterior cruciate ligament tear	Controversial	Yes
Osteophyte	Yes ^b	Yes

^aCartilage defects only; ^bnot independent of magnetic resonance imaging (MRI) changes when examined in the same study

Inflammation is an important factor in knee pain and knee OA pathogenesis

1. Systemic inflammation is associated with knee pain and predicts progression in OA

OA is generally thought of as a ‘non-inflammatory’ type of arthritis; however, localised low-grade inflammation is now known to be an important factor in OA pathogenesis. Pro-inflammatory cytokines, including interleukin (IL)–1 β , tumour necrosis factor– α (TNF– α), and IL–6 are produced by synovium and chondrocytes, and contribute to the progression of cartilage degradation⁹. The inflammatory pattern is observed both early¹⁰ and late⁹ in the disease process.

This local inflammation can be observed systemically using assays for high sensitive C reactive protein (hsCRP), as hsCRP levels are positively correlated with the degree of synovial inflammatory infiltration in OA¹¹. A meta-analysis from our centre demonstrated that hsCRP levels are modestly, but significantly, elevated in persons with OA compared to controls¹². Importantly, hsCRP levels are associated with both symptoms and disease markers, with higher hsCRP levels associated with greater pain (in both this meta-analysis¹², and longitudinal data from our centre^{12, 13}), decreased physical function¹², decreased cartilage volume¹⁴ and disease progression^{10, 15}.

Therefore, **systemic inflammation predicts both pain and structural outcomes in OA**.

2. Local inflammation (effusion–synovitis) is associated with pain and predicts progression of OA

Effusion (excess synovial fluid within the joint space) and /or synovitis (thickening of the synovium), represent localised inflammation within the knee joint. This can be visualised using magnetic resonance imaging (MRI), a highly sensitive method of detecting and measuring knee structural abnormalities. Effusion–synovitis is seen on T2 MRI as increased signal intensity in the joint spaces (see Figure 1).

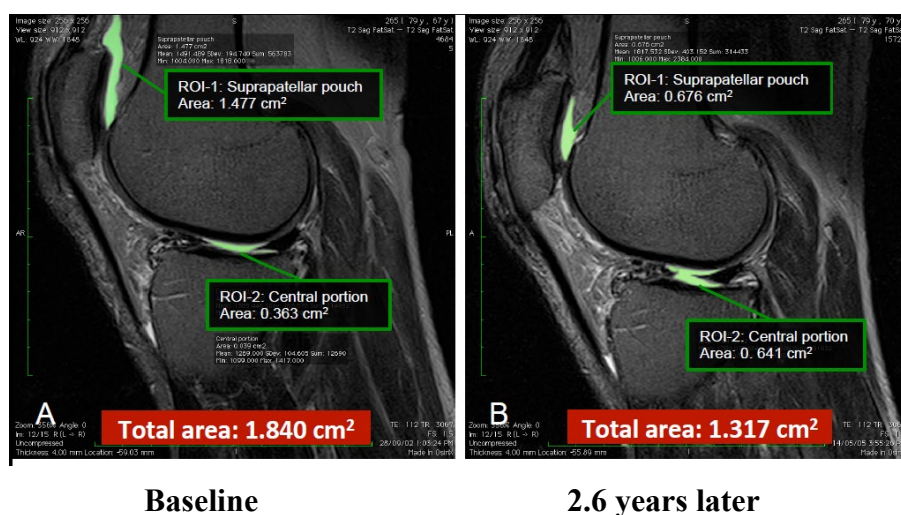


Figure 1: Example of changes in effusion-synovitis area (cm²). Effusion-synovitis size decreased from baseline to follow-up 2.6 years later (A-B). Effusion / synovitis are present in both the suprapatellar pouch and central portion. Overall, effusion size is smaller at follow-up MRI.

Like hsCRP, inflamed synovia and/or effusions are important because they predict pain¹⁶⁻¹⁹, especially non-weight bearing knee pain¹⁹. Importantly, such associations are localised²⁰, with effusions having independent compartment-specific associations with knee pain¹⁹. Our pilot data from the TASOAC study shows that this is common, affecting 67% of randomly selected participants aged ≥ 50 in Hobart²¹, including 22% with large effusions (effusion-synovitis score ≥ 3).

Our data demonstrates that having effusion-synovitis at baseline was more common amongst those who went on to develop new knee pain, or whose pain worsened over 2.6 years [to be presented as an oral presentation at *Osteoarthritis Research Society International*, Seattle, 2015]. Having effusion-synovitis in three of the four anatomic regions in the knee was associated with presence of non-weight bearing pain (Figure 2); and the proportion of study participants with increased knee pain over time was larger in those with larger areas of size of effusion-synovitis on MRI scans (Figure 3). Links between synovial inflammation and knee structural changes are a vicious cycle, with synovial inflammation causing knee structural changes²², and knee structural changes causing synovial inflammation^{22, 23}.

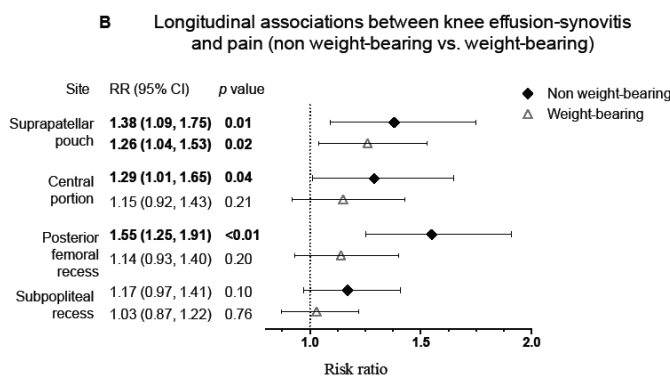


Figure 2: Associations between knee effusion-synovitis in different subregions and increases in total knee pain over 2.6 years in the TASOAC study [unpublished data].

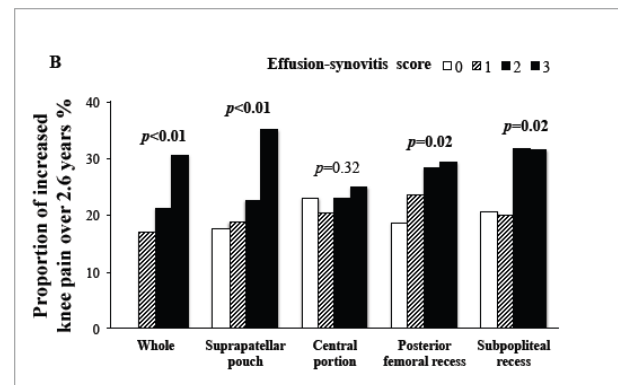


Figure 3: Associations between knee effusion-synovitis in different subregions and increases in total knee pain over 2.6 years in the TASOAC study [unpublished data]

Therefore, **effusion-synovitis is associated with increased knee pain** over time in a dose-response, compartment-specific, manner, and **represents localised inflammation**. Stopping the cascade of inflammation is likely to slow down deleterious changes in knee structure.

Effusions / synovitis predict cartilage loss and joint replacement; therefore, treating these may slow disease progression

Data from our centre (Menzies, Hobart – TASOAC study) recently published in the top journal of the field also demonstrates that effusion-synovitis in the suprapatellar pouch predicts cartilage defects, bone marrow lesions (BMLs) and increased cartilage loss over 2.6 years²¹ (See Figure 4). This is consistent with data from arthroscopy²², and with other longitudinal data that demonstrates that effusion predicts joint replacement²⁴. Thus, **reversing inflammation** has the potential to improve outcomes globally in knee OA.

High levels of inflammation in OA can be treated, but existing options have side effects

A recent RCT in knee OA demonstrated that elevated levels of systemic inflammation can be modified²⁵, with oral low-dose prednisolone reducing hsCRP in people with elevated hsCRP and clinical synovitis by as early as 6 weeks. However, prednisolone is associated with a well-characterised suite of adverse events in the short and long term, and therefore such treatments are not feasible for all people with OA and systemic inflammation. Inflammation can also be reduced using treatment with biological therapy e.g. adalimumab, but these therapies are expensive, and have too many side effects for widespread use. Cyclooxygenase inhibitors e.g. Celebrex are labelled non-steroidal anti-inflammatory drugs (NSAIDs), yet do not change CRP levels²⁶, casting

doubt on their ability to influence inflammation. Therefore, **high hsCRP levels are amenable to change, but safer treatments for reducing inflammation are needed.**

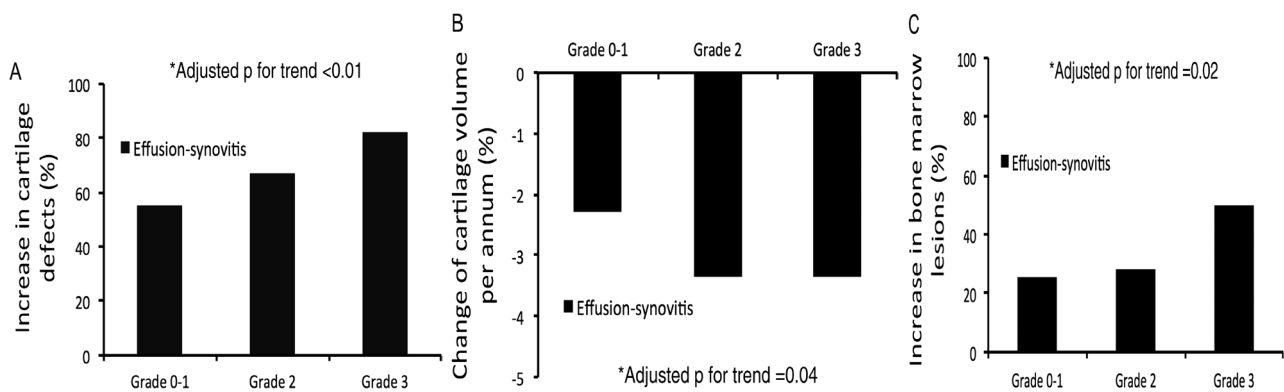


Figure 4: Associations between joint effusion-synovitis in the suprapatellar pouch and (a) cartilage defects (b) cartilage volume (c) bone marrow lesions over 2.6 years. ²¹

Fish oil reduces systemic inflammation in rheumatoid arthritis but not osteoarthritis

Fish oil has been successfully used for treating rheumatoid arthritis (RA) since the early 1990's. The omega-3 fatty acids, eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) can suppress synthesis of the omega-6 pro-inflammatory eicosanoids, prostaglandin E₂ and leukotriene B₄²⁷. Overall, RCT's in RA demonstrate that fish oil reduces pain, morning stiffness, number of painful and/or tender joints and NSAID consumption²⁸; and is an effective add-on therapy to standard RA therapies, reducing risk of treatment failure and increasing rate of remission²⁹.

Therefore, fish oil is an effective treatment for RA.

However, data on efficacy of fish oil in OA is limited. Our Network conducted a 24-month RCT of high dose fish oil vs low dose fish oil / canola oil in participants with knee OA (CIC Hill, *Annals of the Rheumatic Diseases*, under revision). Intensity of knee OA symptoms reduced more in the low dose group at 24 months; suggesting that **fish oil is ineffective for OA, but canola oil may be effective.**

Krill oil is a better alternative for osteoarthritis than fish oil

Krill oil, which is extracted from Antarctic Krill (*Euphausia Superba*, a zooplankton crustacean) has emerged on the market in recent years. Like fish oil, it is high in EPA and DHA³⁰, although the chemical structures of the fatty acids differ (phospholipids, rather than triacylglycerol or fatty acid ethyl esters); but unlike fish oil, it also naturally contains antioxidants (predominantly astaxanthin³⁰). Additionally, bioavailability of krill oil is better than fish oil, as comparable amounts of EPA and DHA are obtained from lower doses of krill oil compared to fish oil³¹.

In animal studies, krill oil supplementation had larger effects on most clinical outcomes than fish oil³² while krill oil reduced the severity of inflammatory arthritis in mice by 50% compared to controls³². In mice transgenic for human TNF- α , both fish and krill oil improved plasma cholesterol levels, but only krill oil had additional beneficial effects on markers of fatty acid oxidation³³. Therefore **bioavailability data and animal studies suggest that krill oil has anti-inflammatory and anti-oxidative effects, and may be a better treatment in vivo than fish oil.**

Pilot data in humans: Deutsch et al³⁴

Only one RCT has assessed the effect of krill oil on inflammation and arthritis symptoms³⁴. This RCT included 90 participants aged 30–75 with cardiovascular disease, RA or OA, and raised CRP levels. Participants treated with 300mg krill oil over 30 days had less pain on the Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) pain scale (change scores -38.35 $\pm$ 21.06 [krill oil] vs -0.6 $\pm$ 15.89 [placebo], $p=0.01$), and reductions in CRP levels over 30 days (-30.9% $\pm$ 1.0 vs +25.1% $\pm$ 1.05, $p=0.008$), compared to placebo. Similar effects were observed with stiffness and functional impairment scores. These results demonstrate that krill oil begins to work by 4 weeks,

and potentially as early as 14 days. ***This suggests that krill oil reduces knee pain and systemic inflammation, within 2-4 weeks.*** No adverse events were reported.

The Deutch et al.³⁴ trial is an important proof of principle study. It demonstrates improvements in pain, stiffness, functional impairment, and systemic inflammation. However, there were limitations.

1. The follow up period was short (30 days).
2. No imaging data was collected; therefore, effect of krill oil on structural changes is unknown.
3. The study population was a mixture of people with disorders that increase systemic inflammation (cardiovascular disease, RA and OA), and no subgroup results were described. There were numerous exclusion criteria that were not justified by contra-indications to krill oil, including any concurrent medical condition or concomitant treatments, limiting the generalisability of the study findings.
4. Adverse events were poorly documented.
5. The study was not independent, being funded by a pharmaceutical company.
6. The dose of krill oil used in this trial (300mg/day) is much lower than that currently available over-the-counter in Australia (and hence used by consumers), as well as being much lower than the dose recommended by experts as being truly anti-inflammatory (2-3 g/day) (personal communication, Trygve Bergeland, Aker Biomarine).

Therefore, we plan to conduct a six month RCT of krill oil 2g/day (vs placebo) to assess the effect on knee pain and size of knee effusions in people with at least moderate knee pain and knee effusions on MRI. This will target an inflammatory subtype of OA with a treatment that has been shown to have anti-inflammatory effects.

This is the first trial to measure change in effusion-synovitis; therefore this approach is cutting edge. Measurement of these features is now possible due to advances in MRI technology and the expertise of our Network.

Research Plan

Study aim

To compare, using a randomised, placebo-controlled double-blind design over six months, the effect of 2g krill oil daily compared to identical placebo, on knee pain (primary outcome) and also MRI-detected knee effusion in 260 participants with clinical knee OA, significant knee pain and effusions on imaging.

Study design

Multi-centre randomised, placebo-controlled double blind clinical trial.

Primary outcome

Change in knee pain (assessed by visual analog score (VAS)) over 24 weeks.

Secondary outcomes

1. Change in maximal area and volume of knee effusion-synovitis, 24 weeks.
2. Change in ordinal measures of knee effusion, 24 weeks.
3. Change in BML maximal area, 24 weeks.
4. Change in ultrasound parameters, 24 weeks
5. Change in knee pain (VAS score), other time points (4, 8, 12, 16, 20 weeks).
6. Change in knee pain (total WOMAC pain, weight bearing and non-weight bearing pain), all time points.
7. Change in hand and back pain (VAS), all time points.
8. Change in knee function, all time points.
9. OMERACT-OARSI responder criteria³⁵, all time points.
10. Change in hsCRP, 12 and 24 weeks.
11. Change in blood lipids, 12 and 24 weeks.

12. Change in blood glucose, 12 and 24 weeks.
13. Leg strength, as measured by dynamometer, 12 and 24 weeks.
14. Analgesic use, all time points.
15. Quality of life (AQoL), all time points, 12 and 24 weeks.
16. Pressure pain threshold testing (PPT) on study knee, 24 weeks
17. Adverse events, 12 and 24 weeks.
18. Cost data, (time points TBA)

Study participants

We will recruit 260 participants from five centres (Hobart, Melbourne, Sydney, Adelaide and Perth) with significant knee pain and presence of MRI defined effusion–synovitis. We will use a proven strategy, including collaboration with general practitioners, specialist rheumatologists, orthopaedic surgeons, and advertising through TV, newspaper and social media. This is the approach we have taken to successfully recruit participants for large community–based trials across a range of musculoskeletal conditions.

Inclusion criteria

1. Aged over 40 years old.
2. Men and women with significant knee pain on most days (defined as a VAS >40mm).
3. Knee effusion–synovitis on MRI: In TASOAC the proportion of participants who had effusion–synovitis on MRI was high (67%), suggesting that localised knee inflammation is common, and that MRI is an appropriate screening tool.
4. Meet American College of Rheumatology (ACR) clinical criteria for knee OA confirmed by a rheumatologist.

Exclusion criteria

1. Inability to have an MRI (claustrophobia, metal in eyes).
2. Severe knee OA (joint space narrowing (JSN) on X-ray of Grade 3 using the Osteoarthritis Research Society International (OARSI) atlas³⁶).
3. Use of anticoagulants, high dose aspirin or non-steroidal anti-inflammatory drugs (NSAIDs), as krill oil is contra-indicated in such people.
4. Unwillingness to stop taking krill oil and fish oil medications during the trial.
5. High dietary fish intake (≥ 1 fish / seafood meal each week)
6. Other forms of inflammatory arthritis (especially rheumatoid arthritis and gout).
7. Seafood allergy.
8. Significant knee injury within the last 6 months.
9. Arthroscopy or open surgery in the index knee in the last 12 months, or planned.
10. Injections of corticosteroids (last 3 months) or hyaluronic acid (last 6 months) in the index knee.
11. Pregnancy or breastfeeding.
12. Use of any investigational drug(s) and/or devices within 30 days prior to randomisation
13. Presence of any serious medical illness that may preclude 24 week follow up.
14. Inability to provide informed consent.

Treatment

Eligible participants will receive two capsules of krill oil or inert identical placebo daily for 24 weeks. Allocation of participants to active or placebo will be based on computer generated random numbers, by site, using adaptive allocation. Allocation concealment will be ensured by using identical capsules, created for us by a compounding pharmacy. Use of a centralised, secure allocation procedure will ensure allocation data cannot be accessed or influenced by any person.

Outcome measures

During the 24 week study, there will be 4 study visits (screening, baseline / week 0, weeks 12, 24) – see Table 2. Additional questionnaires will occur at weeks 4, 8, 16 and 20. MRI's will occur at screening, and week 24. Knee X-ray will occur only at screening, for classification purposes.

1. Primary outcome: Pain

Pain and function will be assessed at each time point using a 100mm visual analogue scale (VAS) and asked “On this line, how would you rate your pain today”? We will also assess pain and function using WOMAC³⁷.

2. MRI measure of change in effusion size

An MRI scan of the “study” knee will be performed (screening, week 24). Knees will be imaged in the sagittal plane on a 1.5T whole-body magnetic resonance unit (GE-Signa) using a dedicated 8-channel knee coil, using a proton density-weighted fat saturation 2-dimensional (2-D) fast spin echo sequence (T2). Effusions are defined as the presence of intra-articular fluid-equivalent signal on T2 MRI sequences. These will be measured using software cursors.

Effusion size in each of 4 regions of interest (ROI; suprapatellar pouch, central portion, posterior femoral recess, subpopliteal recess) will be summed together for each slice. The greatest score for the sum of the four ROI (mm²) will be used as maximal effusion. Intra-observer repeatability for this measurement method is excellent, with an ICC of 0.81. This is superior to an ordinal measure of effusion-synovitis size, with kappa 0.63–0.75 in different sub-regions [in-house data].

Effusion volume will be measured in a similar way but with volumes for each slice summed over the whole joint. Change in effusion-synovitis volume and area is calculated by subtracting baseline values from follow-up maximal values.

Differentiating effusions from synovitis on MR images requires administration of a contrast agent (gadolinium), which may cause nephrotoxicity. While this is rare, we do not consider the benefits of such differentiation between synovitis and effusion justifies the risk to our study participants.

Other explanatory measures

3. Radiographic OA

A standing anteroposterior semiflexed radiograph of the ‘study knee’ will be performed at screening. X-rays will be scored for joint space narrowing and osteophytes on a four point scale (0–3) using the OARSI atlas³⁶. In our hands this method has very high reproducibility with an ICC of 0.98 for joint space narrowing and 0.99 for osteophytes³⁸.

4. Bone marrow lesions present on MRI

BMLs will be assessed on the MRI scans, and defined as an ill-defined hyperintensity in the subchondral bone. One trained observer will score lesions on all MRIs at each time point; they will be blinded to treatment allocation and clinical data. The maximum size of the lesion will be measured in mm² using software cursors applied to the greatest area of each lesion, as previously described^{8,39}. Intraobserver repeatability for this measurement method is excellent, with ICC of 0.97³⁹.

5. Ultrasound parameters

Presence and depth of effusion, synovitis score, presence of synovial hypertrophy and synovial power Doppler signal at the suprapatellar pouch (SPP), lateral and medial para patellar pouches will be assessed. There is very limited data on ultrasound parameters vs MRI in knee OA, therefore this data will add to the field.

6. Blood markers: Inflammation, lipids, glucose

Markers of systemic inflammation, including hsCRP, IL-6 and TNF- α , cholesterol (HDL, LDL), and fatty acids will be assessed using standard assays, from blood samples taken at screening, week 12 and week 24. Changes in hsCRP levels should be detectable within 7 days of krill oil

treatment³⁴. Previous studies show that krill oil reduces lipid levels in people with hyperlipidaemia⁴⁰. As one study has suggested that krill oil may reduce insulin sensitivity⁴¹, we will monitor fasting blood glucose levels at each blood sampling.

7. Lower limb muscle strength

Lower limb muscle strength is a key correlate of pain and will be measured by dynamometry in the lower limbs (involving both legs simultaneously) at weeks 0, 12 and 24. This technique mainly measures quadriceps and hip flexors; these are the most important muscle groups for knee OA based on our studies and those of others; strength in other muscle groups eg hamstrings are not associated with knee OA⁴². Repeatability estimates (Cronbach's α) were 0.91⁴³. Dynamometer-assessed leg strength improves longitudinally when pain reduces (unpublished data, 4Jointz trial)⁴⁴.

8. Anthropometry

These include height (stadiometer) and weight (electronic scales) measured at weeks 0, 12 and 24.

9. Concomitant medication

Use of pain medicines will be recorded by questionnaire at all visits, including over-the-counter remedies, interventions, or supplements. Participants will be asked to keep medications as stable as possible (including NSAIDs) and use paracetamol as rescue medication, which will be provided by us. Use of fish and krill oil during the trial will not be permitted, and use must cease 4 weeks prior to randomisation. Participants who commence anti-coagulant therapy will be withdrawn from the trial.

10. Compliance estimates

Participants will be asked to bring unused softgels with them to their clinic visits at 12 and 24 weeks. These will be counted, enabling estimates of treatment compliance. This is the best method⁴⁵, and is industry standard for measuring compliance.

11. Joint replacement

We plan to collect data on joint replacement, including date of surgery, by asking about knee surgery at each time point, and also by retaining contact with study participants once the study is complete. We will also report time to joint replacement if there is sufficient data. This will be able to be collected both from the patient themselves and from the national joint replacement registry using linked data.

12. Cost data / quality of life

The Assessment of Quality of Life (AQoL) questionnaire can be used to calculate utility values. Data on medication use, adverse events and time away from work will be used as key inputs for health economic evaluations. Cost-effectiveness analysis on the primary outcome measure and cost-utility analysis with the AQoL will be possible if the primary endpoints are met. Cost data and service use will be able to be assessed from linked data, including PBS and MBS data.

13. Service use / medication use

We plan to assess use of services, medications and other aspects of health by linking to datasets available for linkage. These include Medicare and Pharmaceutical Benefits Scheme (PBS) datasets, the National Death Index, the National Joint Replacement Registry, and state-based hospital emergency databases and hospital inpatient databases, and other health datasets.

14. Pressure pain threshold testing

Pressure pain threshold testing (PPT) assesses sensitivity to pain. There is increasing evidence that some people with OA have abnormal sensation in the areas of their body that are painful, including pressure-induced pain⁴⁶. Presence of abnormal sensation of this type has implications for response to pharmaceutical treatment, hence will be assessed where facilities permit (Melbourne, Perth). PPT will be conducted using a pressure algometer (FDN200, Wagner Instruments, USA), applied over 3 test sites: the affected joint (knee, assessed over the patella), an unaffected joint (ipsilateral radioulnar joint), and over the thenar eminence⁴⁷. Fourteen-day test-retest reliability for PPT was

0.85–0.90 (intraclass coefficients)[ref]. PPT will be assessed at the same time points as leg strength (baseline, 12 and 24 weeks).

14. Adverse events

Drug safety will be assessed by recording all important adverse events (AE's), whether we think they are related to the study medications or not, as per the relevant extension of the CONSORT statement⁴⁸. Blood samples will be taken for assessing change in lipids and inflammatory markers (screening, week 12), although we expect these to improve, rather than worsen, with krill oil treatment. We will report incidence and number of regular AE's, by treatment group, and categorised by type. We have no preconceived idea of what these other types might be, except for those mentioned above. We will report incidence and number of serious adverse events (SAE's), including cancer, unplanned hospital admissions and joint replacements. Krill oil has a good safety profile³¹, and is available over-the-counter, so a data safety monitoring board is not required.

Schedule of assessments**Table 2: Schedule of assessments**

Visit / week number	Screening	Baseline (week 0)	4	8	12	16	20	24
Informed consent	x							
Knee x-ray	x							
Knee MRI	x*							x
Clinic measures								
Bloods (fresh / storage)	x				x			x
Clinical examination	x							
Leg strength		x			x			x
Height and weight		x						x
Capsules given		x			x			
Capsule count					x			x
Pressure pain testing (PPT: Mel / Perth)		x			x			x
Ultrasound (Hob / Adl / Perth)		x						x
Questionnaire measures								
Knee VAS	x	x	x	x	x	x	x	x
Knee WOMAC		x	x	x	x	x	x	x
Knee ICOAP		x	x	x	x	x	x	x
Patient global evaluation		x	x	x	x	x	x	x
Joint replacement surgery	x				x			x
Concomitant medications	x				x			x
Safety (AEs)					x			x
Hand VAS, back VAS		x	x	x	x	x	x	x
AQoL		x			x			x
Patient health questionnaire (PHQ-9)		x						
Pain at other sites		x			x			x
painDETECT		x						
Adverse events	As required							
Early withdrawal	As required							

* if otherwise suitable

Statistics

Linear regression will be used to compare changes in pain, maximal effusion size and function in univariate and multivariable modelling (if groups are not well matched at baseline) between groups using intention to treat analysis for all outcomes. Per protocol analyses will be completed for study participants consuming $\geq 80\%$ of capsules. Significance will be $p < 0.05$ (two tailed). Missing data will be accounted for by using propensity weighting, as appropriate.

Power calculations

Based on in-house data from the 4Jointz trial⁴⁴, assuming 10mm difference between krill oil and placebo on the VAS pain scale (reduction in VAS pain scores in the placebo group by -15.5 ± 25.5 mm over 12 weeks), 90% power, and 5% probability of type 1 error ($\alpha=0.05$), we will need 234 participants. Adjusting for 10% loss to follow up, we need 260 participants (130 in each arm). While studies suggest that a 15 mm difference between groups is clinically meaningful,⁴⁹ for a natural therapy which has a better safety profile than pharmaceutical products, lower benefits are also likely to be worthwhile.

Based on data from TASOAC, this will give us 96% power to detect a difference in effusion size of 20% (mean 2.24cm^2 , SD of change 1.35). What is clinically significant has not been defined, but we will investigate associations between change in effusion size and change in pain in this trial. We expect to see a natural fluctuation in knee effusion, but this effect is likely to be similar in intervention and treatment groups. Any differences in effusion size between treatment vs placebo groups can then be explained by the treatment, if the randomisation has worked as planned.

If we use data from the pilot study³⁴, assuming a 20% relative difference in WOMAC scores between groups and 90% power, and 5% probability of type 1 error ($\alpha=0.05$), we will need a total of $n=54$ participants over 4 weeks. However, we believe these figures are overly optimistic, as participants in this trial were required to cease NSAIDs, and the trial had a number of potential biases. We propose to conduct an add-on trial to usual practice rather than treatment cessation, as this reflects the real world more accurately.

Timelines

	2016				2017				2018			
Ethics and study set-up												
Recruitment, screening, and initiation of krill oil												
12 week clinic (pain, function, safety, new capsules)												
24 week visit for MRI scan, pain and function												
Analysis and write up												

Outcomes and significance

OA is a major public health problem that is poorly managed, leaving patients in pain while the disease progresses. Krill oil is cheap and safe, and already widely available in Australia. It appears to successfully treat localised inflammation; therefore, it has excellent prospects for treating an inflammatory subtype of knee OA. If this trial is positive, there will be compelling evidence that krill oil will improve pain and function, change disease trajectory and slow disease progression, in a field that currently has *no approved therapies* for slowing disease progression, which will be easily translatable into clinical practice.

A. References

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